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Trapezoidal Rule:
The Trapezoidal Rule is a simple and effective method for approximating a definite integral of a function $f(x)$ over an interval $[a, b]$. It approximates the area under the curve by dividing the interval into small segments and approximating the area under each segment using trapezoids.
Steps:

- Divide the interval $[a, b]$ into n equal subintervals, each of width $h = \frac{b-a}{n}$.
- Calculate the function values at the boundaries of these subintervals: $f(x_0), f(x_1), ..., f(x_n)$ where $x_i = a + i \cdot h$.
- Apply the trapezoidal rule formula to each subinterval:

$$\int_a^b f(x)\,dx \approx \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + ... + 2f(x_{n-1}) + f(x_n)\right]$$

- Simplify the formula to get the numerical approximation of the integral.

Problem
Consider the table:

x	1	2	3	4	5	6	7
$f(x)$	961.5	466.7	662.8	620.1	757.2	807.6	761.1

Using the Trapezoidal Rule, find $I = \int_1^7 f(x)\,dx$.

1. Divide the interval $[1,7]$ into equal subintervals:

$$h = \frac{7-1}{6} = 1$$

2. Calculate the sum of function values excluding the endpoints:

$$\sum_{i=2}^{6} f(x_i) = f(2) + f(3) + f(4) + f(5) + f(6)$$

$$= 466.7 + 662.8 + 620.1 + 757.2 + 807.6$$
$$= 3314.4$$

3. Apply the Trapezoidal Rule formula:

$$I \approx \frac{h}{2}[f(a) + 2\sum_{i=2}^{6} f(x_i) + f(b)]$$

Where $a = 1$ and $b = 7$.

$$I \approx \frac{1}{2}[f(1) + 2 \cdot 3314.4 + f(7)]$$

$$I \approx \frac{1}{2}[961.5 + 2 \cdot 3314.4 + 761.1]$$

$$I \approx \frac{1}{2}[961.5 + 6628.8 + 761.1]$$

$$I \approx \frac{1}{2} \cdot 8351.4$$
$$I \approx 4175.7$$

Therefore, using the Trapezoidal Rule, the approximate value of $I = \int_1^7 f(x)\,dx$ is 4175.7